

Q7

A) Codd's relational operators

- a) Codd's relational operators lead to closure, which means that the result of a query on a relation is also a relation of the same type. This is advantageous because it allows us to chain multiple relational operators together to perform complex data processing tasks. For example, we can apply a selection operator to a relation to filter out unwanted data, and then apply a projection operator to the resulting relation to retrieve only the desired attributes.

B) Illustrate the four datatypes and list operations

- a) Vector:
 - i) Closure-preserving operation: Vector addition
 - ii) Non-closure-preserving operation: Scalar multiplication
- b) Matrix:
 - i) Closure-preserving operation: Matrix addition
 - ii) Non-closure-preserving operation: Matrix inversion
- c) Complex number:
 - i) Closure-preserving operation: Complex addition
 - ii) Non-closure-preserving operation: Complex division
- d) Color (with RGB components):
 - i) Closure-preserving operation: Color addition
 - ii) Non-closure-preserving operation: Color subtraction